

Guardian Angel

An autonomous, proportional scam-defence system that protects customers **before** irreversible loss, then turns every confirmed case into institutional memory — without adding friction to ordinary banking.

Team **FlowState AI** Category **Intelligent Banking (OCBC)** PolyFinTech100 API Hackathon 2026

1 • THE PROBLEM & INDUSTRY GAP

Singapore lost **S\$1.1 billion to scams in 2024**, with authorised push-payment (APP) scams — where the victim is logged in and authorises the transfer themselves — the fastest-growing category. *(Singapore Police Force, Annual Scams & Cybercrime Brief 2024)* Ordinary authentication cannot stop this: the customer *is* the one clicking confirm. Existing bank controls act mainly at the last step — the transfer itself — but the scam begins far earlier, and victims aged 50+ account for a disproportionate share of losses as Singapore's population ages toward **1 in 4 residents being seniors by 2030**.



Most bank controls watch here ↑ (too late). Guardian Angel reasons across the **whole journey**, and closes the loop after loss occurs at any one customer, not just that one transaction.

2 • PROPOSED SOLUTION

Guardian Angel is an autonomous scam-defence layer for OCBC that reasons over a customer's **caller identity, device behaviour, recipient network history and transaction pattern together** — not the amount alone — to decide the least-intrusive protective action, and automatically opens a case for OCBC's fraud team when it acts. **What makes it different:** a deterministic, auditable rule engine computes risk; a bounded AI second opinion can only ever raise it, never lower it; and a policy engine — not the AI — is the sole enforcer of what happens to the customer's money. When staff confirm a case as a scam, that account is flagged for every future customer OCBC ever asks about it — the system gets smarter with every case, not just faster.

3 • HOW IT WORKS — THREE INTERFACES, ONE RISK ENGINE

Customer App "/"	Staff Console "/staff"	Guardian Contact "/guardian"
Grace's OCBC app — transfers, Verify Caller and Adaptive Care folded into one screen. Never shows a score; only a plain-language ask.	OCBC risk-ops login → dashboard → case log → case investigation → Risk Account Registry . Where a human closes the loop.	The nominated guardian (e.g. an adult child) — informed, never in charge . Read-only alerts, no approval power over the account.

1 Caller Verification A live in-app code the caller must read back confirms a genuine authenticated OCBC session — stops impersonation before it starts.	2 Recipient Network Check Screens the recipient for mule-account patterns and, where a contact email is supplied, a real breach-exposure lookup.
3 Smart-Lock (Money Lock) Detects account-takeover sequences (new device + credential reset + limit change) and secures funds — reversible, never frozen.	4 Adaptive Financial Care Detects sustained manipulation over weeks and escalates proportionally: info → questions → cooling-off → guardian → human support.

THE RISK ENGINE — NOT ONE SIGNAL, AN ADAPTIVE COMBINATION
Every signal is tagged to a **family** (transaction, behavioural, device-security, recipient-network, identity). A **diversity gate** requires evidence from **2+ independent families** before a high-risk action is allowed to stand — **a large amount alone is never high risk**. A bounded AI second opinion (Claude) can only **raise** the deterministic score, never lower it; a separate **policy engine enforces the action and can never block funds directly** — every decision reproducible, auditable and explainable.

- R1 **Account-Takeover** — 2+ security changes + balance drain
- R2 **Scam-Recipient** — large + new overseas payee + network flag
- R3 **Social-Engineering** — new payee + savings liquidation + dismissed warnings
- R4 **Compromise + Mule** — new device + network-flagged recipient

NEW — RISK ACCOUNT REGISTRY & THE STAFF FEEDBACK LOOP
Every transfer's recipient account is cross-referenced against a Risk Account Registry (Blacklisted / Watchlist / Cleared) — a real signal feeding the score, not a static list. When a case scores MEDIUM or above, it **auto-opens** in the Staff Console; when staff confirm it as a scam, the registry updates and **every future transfer to that account is scored differently**, for any customer.

AI detects → Case opens → Staff confirms → Registry updates → Next transfer cross-referenced

4 · MARKET IMPACT & USE CASES

TAM

\$S\$1.1B

Annual scam losses across Singapore, 2024 — the addressable harm a transaction-layer defence can reduce. (*SPF 2024*)

SAM

~4.5M

OCBC's Singapore retail & wealth customers exposed to APP-scam risk on every transfer they make. (*OCBC Group, public disclosures*)

SOM

~1M+

Phase-1 target: OCBC customers aged 55+ and their nominated guardians — the highest-risk cohort for APP and long-con scams.

Grace, 68 — retail customer

A caller claiming to be "OCBC Fraud" pressures her toward a S\$15,000 transfer to a new payee right after a fixed-deposit payout. Guardian Angel verifies the caller fails, flags the recipient, and holds the transfer with a plain-language reason — Grace stays in control and can release it herself in seconds if it's genuine.

Wei Ming — OCBC risk-ops analyst

Grace's held transfer auto-opens a case in his Staff Console with the exact signals, matched pattern and AI assessment already laid out. He confirms it as a scam in one click — the recipient account is now blacklisted for every OCBC customer, not just Grace.

5 · TECHNICAL FEASIBILITY & STACK

React + Vite + TypeScript + Tailwind (3 apps)

Node.js + Express + TypeScript backend

Anthropic Claude API — claude-sonnet-5

XposedOrNot breach-lookup API

A **working, live prototype** — not a mockup: real forced-tool Claude calls power the AI second opinion and action recommendation, with an honestly-labelled cached fallback (`source: 'live' | 'cached\_fallback'`) if a call times out. The rule engine, diversity gate, policy enforcement, Money Lock, verification-code lifecycle and the staff feedback loop are all real executing code — see the worked example below, traced against the actual scoring weights, not asserted.

Worked example — traced live against the current build, not hard-coded

S\$15,000 to David Lim, a brand-new payee, shortly after a savings liquidation, while his account sits on the Risk Account Registry **watchlist**. Four signals fire across **three families**: *behavioural* (new recipient 18, savings liquidation 30) + *transaction* (amount far exceeds typical, 30) + *recipient-network* (registry watchlist, 18) = **96/100**. Diversity gate passes on family count alone — no pattern rule even needed.

Result: CRITICAL → reversible hold + Money Lock + guardian informed + case opens in the Staff Console automatically. When staff confirm the case, the registry escalates David Lim to **blacklisted**; the identical transfer re-run then scores **100/100**. *By contrast, S\$1 to an unrelated, clean new payee produces one signal in one family — stays LOW, opens no case.*

6 · FINANCIAL & TIMING VIABILITY

Financial viability

- ✓ Marginal cost per assessment ≈ 1–2 Claude API calls — cents, not dollars, per protected transfer.
- ✓ No new customer-facing hardware or channel; runs inside OCBC's existing app and ops tooling.
- ✓ A single prevented S\$15,000 case (this build's own worked example) already covers years of API cost for that customer.
- ✓ Registry-driven learning means detection cost per case **falls** as institutional memory grows — the opposite of a static rules list.

Timing viability

- ✓ **Now:** working 3-interface prototype — User, Staff, Guardian — on a shared backend, demoed live.
- ✓ **Phase 1 (0–6m):** pilot live scam interception with a limited risk-ops cohort.
- ✓ **Phase 2 (7–12m):** vulnerability monitoring at scale + Great Eastern data synergies.
- ✓ **Phase 3 (Y2+):** regional scaling of the registry as a shared industry signal.

ONE SYSTEM, EVERY STAGE OF THE SCAM

Scam type	Primary layer(s)	What stops it
Impersonation (fake OCBC/police)	Caller Verification	Caller can't produce the live in-app code → not verified.
Mule / scam-recipient	Recipient Network + Registry	Recipient flagged as mule-like or already on the Risk Account Registry.
Phishing / takeover	Smart-Lock	Device + credential + limit changes → funds secured in Money Lock.
Romance / long-con	Adaptive Care	Sustained escalation over weeks → proportionate, human-supported.
Repeat offender accounts	Staff feedback loop	One confirmed case protects <i>every</i> future OCBC customer from that account.

Guardian Angel moves banking from "Is this transaction suspicious?" to "Is this customer being manipulated — and what did we learn that protects the next one?"